

ADS - PyQ

Quiz 2

Let X be a multivariate random vector in $\mathbb{R}^d$ whose mean distance from the origin is 10. Let p be the probability that the distance of X from the origin is at most 50.

Which of the following is true? Enter the tightest bound possible.

OPTIONS :

- ☒ $p \geq 0.8$
- ☐ $p \leq 0.8$
- ☐ $p \geq 0.2$
- ☐ $p \leq 0.2$

$$p = P(X \leq 50)$$

$$P(X \geq a) \geq 1 - \frac{E[X]}{a}$$

$$E[X] = 10$$

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$p \geq 1 - 0.2 \geq 0.8$$

Consider the JL lemma applied to two datasets, D_1 and D_2 , both having the same number of data-points for a value of $\epsilon = 0.01$. The data-points in D_1 live in $\mathbb{R}^{d_1}$ and those in D_2 reside in $\mathbb{R}^{d_2}$. The ratio $\frac{d_1}{d_2} = 1000$. Let k_1 and k_2 be the dimensions of the lower-dimensional spaces respectively. Choose the most appropriate option.

OPTIONS :

- ☐ k_1 is at least $1000 \cdot k_2$
- ☐ k_2 is at least $1000 \cdot k_1$
- ☐ k_1 can never be equal to k_2
- ☒ k_1 and k_2 can be chosen to be equal to each other

$$\begin{matrix} D_1 \in \mathbb{R}^{d_1} \\ D_2 \in \mathbb{R}^{d_2} \end{matrix} \quad \frac{d_1}{d_2} = 1000$$

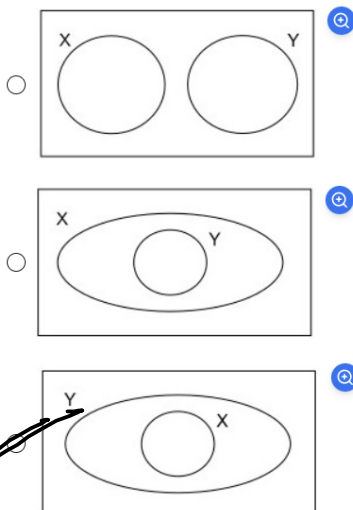
$$k \in O\left(\frac{\log n}{\epsilon^2}\right)$$

no d_1 present

$\therefore k$ is independent of the choice of d .

Which of the following Venn diagrams represent the relationship $X \Rightarrow Y$ for two events X and Y ?

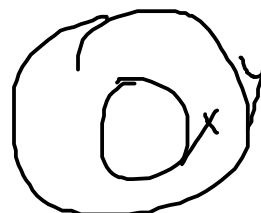
OPTIONS :



$$X \Rightarrow Y$$

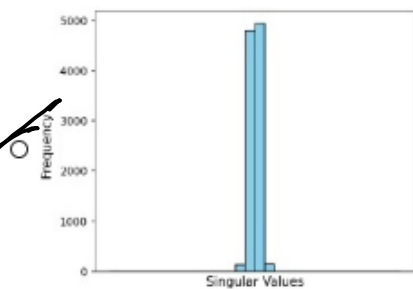
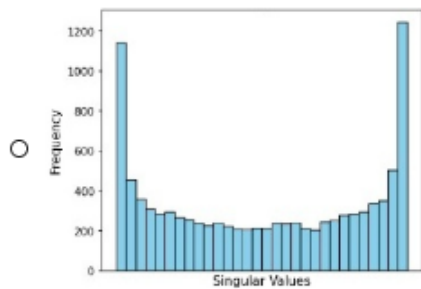
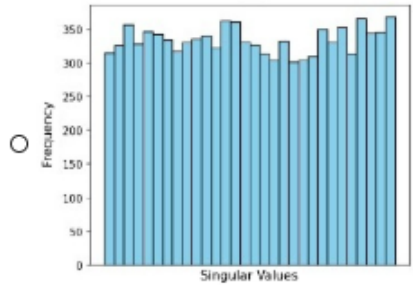
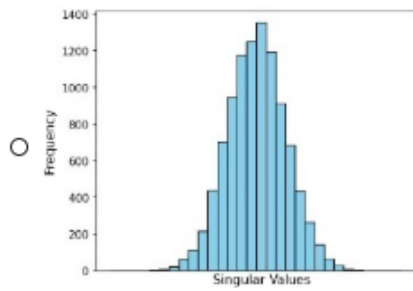
as per the bound

$$P(X) \leq P(Y)$$



Let A be a $10^4 \times 10^5$ random matrix, whose entries are i.i.d standard normal variables. What would the distribution of the non-zero singular values of A look like?

OPTIONS :



$AA^T \Rightarrow 10^4 \times 10^4$
 $(sq.)$
 $\downarrow$
 eigenvalues

π singular value

sq. root of eigenvalues

Let $A = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$. Consider a random variable $\omega \sim \mathcal{N}(0, I_2)$. Find the covariance matrix of $A\omega$.

$A\omega = AA^T$

OPTIONS :

☒ $\begin{bmatrix} 17 & 7 \\ 7 & 10 \end{bmatrix}$

☐ $\begin{bmatrix} 10 & 7 \\ 7 & 17 \end{bmatrix}$

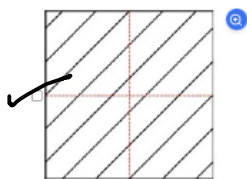
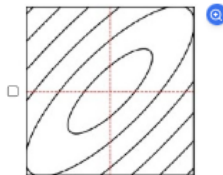
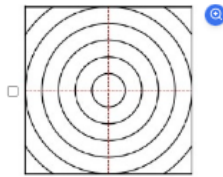
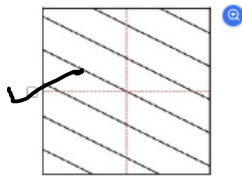
☐ $\begin{bmatrix} 1 & 16 \\ 9 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 7 \\ 7 & 10 \end{bmatrix}$$

Let $\mathcal{H}_w$ be a Datar-style (r, c, p_1, p_2) -sensitive hash family $(h_i : \mathbb{R}^2 \rightarrow \mathbb{Z})$. Suppose we sample a $h \sim \mathcal{H}_2$. In a graphical sense, h partitions $\mathbb{R}^2$ into a countable number of regions, assigning an integer to each region. Which of the following are possible partitionings corresponding to a h sampled from this family?

OPTIONS :



$$w=2$$

$$r, c, p_1, p_2$$

$$h = \left\lfloor \frac{a^T x + b}{a} \right\rfloor$$

$$h = a^T x + b = k$$

$$a \cdot x + b = 2k$$

constant + bias $\Rightarrow$ straight lines

AND
+
OR

Let $\mathcal{H}_w$ be a Datar-style (r, c, p_1, p_2) -sensitive hash family $(h_i : \mathbb{R}^2 \rightarrow \mathbb{Z})$. An LSH is built using L tables, each having k hash functions sampled from $\mathcal{H}_w$. Collision between 2 points happens if any one of the L tables exactly match.

Suppose we increase the number of tables L , while keeping the other parameters constant. Select the correct options:

OPTIONS :

- ☒ True Positive Rate increases $\uparrow$
☒ False Positive Rate increases $\uparrow$
☐ True Positive Rate decreases
☐ False Positive Rate decreases

$$TPR = 1 - e^{-\theta} = 1 - (1 - p_1^k)^L$$

$$FPR = \theta n^{-\beta} = 1 - (1 - p_2^k)^L$$

$$\beta = \frac{\log(1/p_1)}{\log(1/p_2)}$$

$L \uparrow$, TPR & FPR
log. increase

Consider a dataset for an unsupervised learning problem consisting of n points in $\mathbb{R}^d$, where all the points lie on the unit hypersphere. Using the JL lemma, let f be a linear map from $\mathbb{R}^d \rightarrow \mathbb{R}^k$ for $\epsilon = 0.0201$ and a suitable choice of k as mandated by the lemma. Find an upper bound for the distance between any two points of the dataset in the lower dimensional space $\mathbb{R}^k$. Your answer should have exactly two places after the decimal point

Hints

- A hypersphere in $\mathbb{R}^d$ is the set of points with $\|x\| = 1$, where $x \in \mathbb{R}^d$.
- The triangle inequality states that the sum of the lengths of two sides of a triangle is greater than the length of the third side.

Answer (Numeric):

Answer 2.02

$f: \mathbb{R}^d \rightarrow \mathbb{R}^k$ $\epsilon = 0.0201$
all are in the hypersphere $\Rightarrow \|x\| = \|y\| = 1$

$$D_{\max} = 2$$

we have to find the upper bound

$$\|f(x) - f(y)\|^2 \leq (1 + \epsilon) D_{\max}^2 \quad D_{\max} = \|x - y\|$$

$$\|f(x) - f(y)\|^2 \leq (1 + 0.0201) 4$$

$$\leq 1.0201 \times 4 \leq 4.0804$$

$$\|f(x) - f(y)\| \leq 2.02$$

Let A be a matrix whose non-zero singular values are 50, 20, 10, 1, 0.01 and 0.001. If B is the best rank-4 approximation of A , then compute $\|A - B\|_2$.

Answer (Numeric):

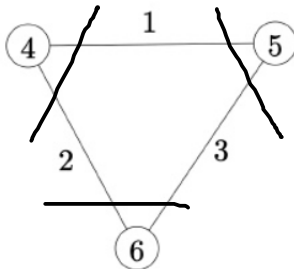
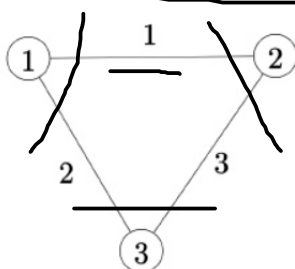
Answer 0.01

$$A (\text{sing.}) = 50, 20, 10, 1, 0.01, 0.001$$

$$B =$$

Consider the following weighted, undirected graph. If an edge is not present in the graph, the corresponding weight is zero.

Based on the above data, answer the given subquestions.



$$\begin{cases} 1 \setminus 2, 3 \\ 2 \setminus 1, 3 \\ 3 \setminus 1, 2 \end{cases}$$

$$1 + 2 = 3$$

$$1 + 3 = 4$$

$$2 + 3 = 5$$

$$\begin{cases} 1, 2 \setminus 3 \\ 1 \setminus 2, 3 \end{cases}$$

$$\begin{cases} (1, 3) + (2, 3) \\ (1, 2) + (1, 3) \end{cases}$$

Question 12 : 6406531337575

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Total Mark : 4.00 | Type : SA

Among all cuts $A \subset \{1, 2, 3, 4, 5, 6\}$ such that $|A| = 1$, find a cut that has the smallest cumulative sum of weights and enter the value of this sum. Note that $|A|$ denotes the cardinality of A .

3 ✓

Find the Laplacian, L , of the graph.

OPTIONS :

☒ $\begin{bmatrix} 3 & -1 & -2 & 0 & 0 & 0 \\ -1 & 4 & -3 & 0 & 0 & 0 \\ -2 & -3 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & -1 & -2 \\ 0 & 0 & 0 & -1 & 4 & -3 \\ 0 & 0 & 0 & -2 & -3 & 5 \end{bmatrix}$

☐ $\begin{bmatrix} 0 & -1 & -2 & 0 & 0 & 0 \\ -1 & 0 & -3 & 0 & 0 & 0 \\ -2 & -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & -1 & 0 & -3 \\ 0 & 0 & 0 & -2 & -3 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} -3 & 1 & 2 & 0 & 0 & 0 \\ 1 & -4 & 3 & 0 & 0 & 0 \\ 2 & 3 & -5 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 1 & 2 \\ 0 & 0 & 0 & 1 & -4 & 3 \\ 0 & 0 & 0 & 2 & 3 & -5 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 2 & 0 & 0 & 0 \\ 1 & 2 & 3 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 & 3 & 3 \end{bmatrix}$

$$L = D - W$$

$$f^T L f$$

$$D = \begin{bmatrix} \sum_{j \neq i} w_{ij} & 0 & \dots \\ 0 & \sum_{j \neq i} w_{ij} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

How many zero eigenvalues does L have?

Answer (Numeric):

Answer

$$6 - 4 = 2$$

$\text{rank}(A) =$ non-zero singular values / linear independent cols
 $= 4$